# ONCE: Boosting Content-based Recommendation with Both Open- and Closed-source Large Language Models

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## Abstract

This study investigates the reproducibility of Liu et al.'s method for using both open-source and closed-source LLMs for recommendation, focusing on validating their approach of enhancing content-based recommendation systems through comprehensive experiments. We leverage the deep layers of open-source LLMs as content encoders to enhance content representation at the embedding level, while for closed-source LLMs, we use prompting techniques to enrich training data at the token level. The experiments are conducted on the EB-NeRD dataset to evaluate the effectiveness of using both open-source and closed-source LLMs for recommendation. The study confirmed the robustness of the original findings, showing slight performance improvements in NAML and NRMS models with the inclusion of additional features, specifically, sentiment labels. However, the ONCE replication unexpectedly did not lead to an overall improvement in performance, which we attribute to issues with enriched tokenization integration.

## 1 Introduction

Personalized content-based recommender systems have become essential tools for helping users navigate the vast and ever-growing amount of information available on digital platforms, such as news websites and book recommendation services. The central element of content-based recommender systems is the content encoder, which encodes the textual data of items to capture their semantic attributes. Historically, recommendation models Wu et al. (2019a) often employed convolutional neural networks (CNNs) as content encoders, usually starting with pre-trained word embeddings like GloVe Pennington et al. (2014). Recent advancements in large language models (LLMs) have shown tremendous potential in various natural language processing tasks due to their deep semantic comprehension and extensive knowledge base gained from pretraining Radford et al. (2018). Previous research has demonstrated the success of integrating LLMs into recommendation systems, such as BERT4Rec Sun et al. (2019) and Transformers4Rec Kang & McAuley (2018), which utilize transformer architectures for sequential recommendation tasks. Moreover, LLMs can be used as a recommender system by predicting an item a user will buy given its history, where you query the LLM with the user's history Li et al. (2023). Liu et al. (2024) explores the innovative use of both open- and closed-source LLMs to enhance the effectiveness of content-based recommendation systems. For closed-source LLMs such as GPT-3.5, Liu et al. (2024) uses prompting techniques to enhance the training data at the token level. Moreover, Liu et al. (2024) employs open-source LLMs like LLaMA to act as content encoders. These models are fine-tuned to improve the representation of content at the embedding level. This work aims to reproduce the idea of Liu et al. (2024) on the dataset EB-NeRD. As a summary, the following contributions are made in this work:

1. Most primary experiments carried out by Liu et al. (2024) have been replicated to assess the extent to which the author's findings can be reliably reproduced on the EB-NeRD dataset.
2. This study takes inspiration from the data-enrichment aspect of the closed-source LLM integration and expands it further by adding sentiment as a feature of the news articles.

## 2 Scope of reproducibility

The work Liu et al. (2024) adresses the problem of enhancing content-based recommendation systems by leveraging both open-source and closed-source LLMs. Traditional content-based recommenders struggle with fully understanding the content due to limitations in capturing deep semantic features. This work proposes a framework called ONCE that aims to improve these systems by integrating the strengths of both open- and closed-source LLMs.

The paper introduces the **ONCE** framework, which leverages both open-source and closed-source LLMs to enhance content-based recommendation systems. The framework employs two strategies:

1. **DIRE (Discriminative Recommendation Framework):** This strategy uses open-source LLMs as content encoders. It fine-tunes these models to enrich content representations and enhance user modeling.
2. **GENRE (Generative Recommendation Framework):** This strategy uses closed-source LLMs (specifically GPT-3.5) to generate enriched data through prompting. This generative approach aims to augment the training data, thereby improving downstream recommendation performance.

The claims the paper make are as follows:

1. Using open-source LLMs as content encoders significantly enriches the representation of content at the embedding level, leading to improved recommendation accuracy. This is evidenced by integrating open-source model LLaMA into the recommendation system and comparing the performance with traditional models.
2. Closed-source LLMs, when employed through prompting techniques, can enrich training data at the token level, enhancing the quality of recommendations. This is supported by employing GPT-3.5 to generate enriched textual and user features and measuring the impact on recommendation performance.
3. The synergistic use of both open-source and closed-source LLMs results in a more effective recommendation system, achieving a significant relative improvement over existing state-of-the-art models. This claim is validated through comprehensive experiments that demonstrate a relative improvement in recommendation quality compared to state-of-the-art models.

## 3 Methodology

The author has publicly shared their experimental implementation on GitHub, where the repository is well-organized and features a comprehensive README file. The file has a very comprehensive overview of the re-training and testing of the DIRE training pipeline. Moreover, the README file includes a overview of the requested code that they call the GPT-3.5-turbo API with to generate the augmented data.

### 3.1 DIRE: Finetuning Open-source LLMs

The **DIRE** leverages open-source LLMs as content encoders to enhance the performance of content-based recommendation systems. This framework integrates advanced embedding techniques and fine-tuning strategies to improve the semantic understanding and representation of content, which are critical for personalized recommendations.

**Embedding layer.** In the framework, natural language templates are utilized for concatenation, replacing the token segmentation approach typically employed by smaller-scale pretrained language models such as BERT. For instance, a news content item with attributes like title, abstract, and category is prefixed with a

label ("news article:") and corresponding tags (<title>, <abstract>, etc.). This transformation consolidates multi-field content into a cohesive individual sequence. The sequence is then mapped into a continuous embedding space of dimensionality $d_n$ using pretrained token embeddings from the LLM. **Attention Fusion Layer.** The LLM architecture, like LLaMA, uses multiple layers of Transformer blocks to sequentially process and refine hidden states, allowing the model to learn complex relationships and capture deep semantic features in the text. To combine the hidden states from the final layer into one clear content representation, DIRE uses an attention fusion layer. This layer converts the high-dimensional hidden states into a smaller $d$-dimensional space using a linear transformation, defined by:

$$\mathbf{Z} = \mathbf{E}^i\mathbf{W} + \mathbf{b} \in \mathbb{R}^{l\times d}, \tag{1}$$

where $W \in \mathbb{R}^{d^n\times d}$ and $b \in \mathbb{R}^d$ are the learnable parameters of the linear transformation. And $\mathbf{E}$ is defined by:

$$\mathbf{E}^i = \text{TransformerLayer}\left(\mathbf{E}^{i-1}\right) \in \mathbb{R}^{l\times d^n}, \quad i \in \{1,\ldots,H\} \tag{2}$$

Then, an additive attention mechanism merges this reduced representation into a single vector:

$$z = \text{Attention}(Z) \in \mathbb{R}^d, \tag{3}$$

This vector is then used in the user modeling or interaction module to help make personalized recommendations. **Finetuning Strategy.** Given the significant computational demands of running LLMs, the lower layers of the LLM, which possess more generalized features, are kept frozen. Only the uppermost layers, which contain task-specific features, are fine-tuned. A caching strategy is adopted to precompute and store the hidden states from the lower layers for all content within the dataset, substantially reducing computational costs. For example, in the LLaMA-7B model with 32 layers, only the top 2 layers may be fine-tuned, reducing the computation load to approximately 6% of the original cost. Low-Rank Adaptation (LoRA) introduces trainable rank decomposition matrices into the pretrained model layers, significantly reducing the number of trainable parameters required for downstream tasks. This approach allows for more efficient fine-tuning, often outperforming traditional methods by drastically decreasing the parameter count. In the DIRE framework, LoRA is applied to the unfrozen Transformer layers, which are the most parameter-intensive components of the model.

### 3.2 GENRE: Prompting Closed-source LLMS

The **GENRE** utilizes closed-source LLMs, to be specific GPT-3.5, to enhance content-based recommendation systems through data augmentation and enrichment techniques. The framework focuses on leveraging the powerful text comprehension and generation capabilities of closed-source LLMs to generate enriched data, improving the overall performance of recommendation models.

In the GENRE framework, closed-source LLMs play multiple roles to enhance the performance of content-based recommendation systems: **LLMs as Content Summarizers.** Closed-source LLMs are proficient at summarizing text content into concise phrases or sentences due to their extensive training on large-scale natural language data. By providing the content title, abstract, and category as input, the LLM generates more informative titles. These enhanced content features replace the original ones during downstream training, improving the content encoder's ability to capture semantic distinctions.

**LLMs as User Profilers.** User profiles typically include preferences and characteristics such as age, gender, topics of interest, and geographic location. Due to privacy policies, such information is often anonymized or missing from datasets. Closed-source LLMs can infer these profiles by analyzing users' browsing history. For example, given a user's news browsing history, the LLM can generate topics and regions of interest, enriching the user profile with inferred data that can be used to enhance collaborative filtering and content recommendations.

**LLMs as Personalized Content Generators.** Recent studies have shown that LLMs can effectively learn from few examples, making them suitable for generating personalized content based on limited

user interactions Dai et al. (2023). For new users with sparse browsing history, the LLM can generate synthetic content pieces that match their inferred interests. These synthetic content pieces help the history encoder better capture user preferences, enhancing the model's ability to provide accurate recommendations even for users with limited data.

**Chain-based Generation.** In this approach, data generated by the LLM is used to refine the dataset iteratively. For example, the LLM first generates topics and regions of interest for a user based on their history. This enriched profile is then used to generate synthetic content that better matches the user's interests. This iterative process allows the model to engage in "chain thinking," producing more contextually relevant and personalized content over multiple iterations.

### 3.3 Experimental Setup

**Dataset.** This study uses the Ekstra Bladet News Recommendation Dataset (EB-NeRD), which is specifically designed for news recommendation research. The large-scale dataset comprises over 2.7 million users and over 600 million impression logs, collected from Ekstra Bladet, a major Danish news outlet. The dataset spans a six-week period from April 27th to June 8th, 2023, a time frame chosen to exclude major events, ensuring the data reflects typical user behaviour.

EB-NeRD is segmented into several components. The *behavior logs* capture detailed user interactions over seven days, including information such as impression timestamps, device types, read times, and scroll percentages. These logs are essential for understanding user engagement and preferences. Accompanying these are the *click histories*, which document users' news consumption over a preceding 28-day period, offering a hindsight view of their reading patterns.

Complementing the user interaction data is a collection of over 120,000 news articles. Each article is enriched with textual features like titles, abstracts, bodies, and additional metadata such as categories, sentiment and publication timestamps. For practical use, the dataset is available in multiple sizes - demo, small, and large - each containing training and validation sets. This structured format enables in systematic debugging and training and testing of recommendation models.

Overall, EB-NeRD provides a rich and clean dataset, ideal for exploring advanced news recommendation methodologies, like the integration of both open-source and closed-source LLMs which this study investigates.

| **Dataset** | **EB-NeRD** |
|---|---|
| | Original |
| # content | 20,738 |
| tokens/title | 6.69 |
| characters/title | 42.0 |
| # users | 18,827 |
| # new user | 368 |
| content/user | 136 |
| content/user$_n$ | 5.0 |
| | **Content Summarizer (CS)** |
| tokens/title | +5,69 |
| characters/title | +39,88 |
| | **User Profiler (UP)** |
| topics/user | 5.83 |
| regions/user | 1.07 |

Table 1: Data statistics. We use "user$_n$" to denote new users. Green numbers signify improvements over the original dataset, while blue numbers indicate the values of newly introduced features.

**Recommendation models.** We assess the efficiency of the proposed ONCE method using recognized content-based recommendation models, namely NRMS, Fastformer, and NAML. The NRMS model Wu et al. (2019b) enhances news recommendations using multi-head self-attention to capture word and news interactions. It has two main parts: the news encoder, which learns representations from news titles, and the user encoder, which captures the relationships between browsed news articles. Both use additive attention to highlight important words and news for more accurate recommendations. The Fastformer model Wu et al. (2021) is an efficient variant that uses additive attention to model global contexts. It transforms the input embedding matrix into query, key, and value sequences, summarizes the query sequence into a global query vector using additive attention, and then models interactions between this global query vector and the keys and values using element-wise products. This approach reduces computational complexity to linearity while effectively capturing contextual information. The NAML model Wu et al. (2019a) makes use of three essential modules ( news encoder, user encoder and click predictor). The news encoder learns news representation by making use of attentive multi-view learning, the user encoder uses attention mechanisms to learn user representations instead. Lastly, the click predictor aims at predicting the probability of a user browsing a news article via inner product product of the representation vectors of user and candidate news.

**Evaluation metrics.** We stick to the standard practice of assessing the performance of news recommendation models using widely accepted metrics, namely, AUC, MRR, and nDCG.

## 4 Results

### 4.1 Results reproducing original paper

Table 2: Performance comparison of various models on the EB-NeRD dataset. (Part 1)

| | **Model** | NRMS (2019c) | | | | Fastformer (2021b) | | | |
|---|---|---|---|---|---|---|---|---|---|
| | **Metric** | AUC | MRR | N@5 | N@10 | AUC | MRR | N@5 | N@10 |
| **DIRE** | LLaMA7B (Ours) | 55.77 | 34.55 | 38.53 | 46.30 | 55.95 | 34.75 | 38.72 | 46.47 |
| **ONCE** | **ONCE (Ours)** | 55.65 | 34.16 | 38.31 | 46.17 | 55.68 | 34.19 | 38.26 | 46.02 |

Table 3: Performance comparison of various models on the EB-NeRD dataset. (Part 2)

| | **Model** | NAML (2019a) | | | |
|---|---|---|---|---|---|
| | **Metric** | AUC | MRR | N@5 | N@10 |
| **DIRE** | LLaMA7B (Ours) | 55.53 | 34.53 | 38.48 | 46.36 |
| **ONCE** | **ONCE (Ours)** | 55.41 | 34.22 | 38.25 | 46.07 |

During the reproduction phase of the original paper, several key factors and adjustments were needed. First and foremost the tokenization process was an instrumental adjustment into enabling the compatibility and smooth training & testing phase w.r.t the already existing building blocks found in the original paper ( Liu (2023a) , Liu (2023b)). During this above-mentioned adjustment, careful adaptation was required during the data reading phase, tokenizer initialization to properly and effectively read/filter news articles and user interactions. In more detail, the correct column selection and filtering from the Eb-NeRD dataset, the usage of multilingual BERT tokenizer for text fields and the handling of negative sampling were all crucial steps in ensuring accurate processing. The usage and reliance on only one open-source LLM (LLaMA-7B) in contrast with the original paper which also made use of a larger LLM (LLaMA-13B) was another adjustment needed to conduct our experiments in a timely manner. More specifically, when trying to train various news recommendation models on the dataset with the 13B parameter version of LLaMA we encountered several OOM (Out-of-Memory) constraints that eventually prevented us from using it. With this being said, we proceeded into making use of the smaller version of LLaMA. Firstly, as seen in table 2 and3, our replication of the DIRE framework showed that the best performing news recommendation model when employing the open LLM (LLaMa-7b) is the FastFormer model. Overall, all of the 4 used metrics and their values are

the most dominant and best performing when tested with the FastFormer model Wu et al. (2021) which can be partially accredited to the architecture and contents of the model itself. More specifically, its usage of additive attention to handle long sequences while handling it with linear complexity is what enables it to efficiently manage the Eb-NeRD data. Moreover, its usage of a global query vector is what enables an efficient interaction between the key vector and the global context-aware key matrix which ultimately ensures an accurate and context-relevant recommendation. Secondly, our ONCE implementation did not lead to an improvement in performance for several reasons: although we successfully implemented enriched tokenization (GENRE) and trained using the DIRE architecture, the new data generated through this process was potentially not effectively integrated into the final training phase. This issue is common and has been reported by other researchers. Our efforts to resolve it, including reaching out to the original author, were inconclusive and further limited by constraints in time and computational resources.

### 4.2 Results beyond original paper

In our study, we extended the experimental setup to include sentiment as an additional feature in the recommendation models. This decision was motivated by insights from the GENRE framework, which demonstrated that enriching the data could enhance performance. We hypothesized that incorporating sentiment labels into the content could provide additional context that might improve the models' understanding and predictions.

Table 4: Performance comparison of various models on the EB-NeRD dataset with sentiment addition.

| | **Model** | NAML (2019a) | | | | NRMS | | | | FastFormer | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Metric** | AUC | MRR | N@5 | N@10 | AUC | MRR | N@5 | N@10 | AUC | MRR | N@5 | N@10 |
| **DIRE** | DIRE (replication only) | 55.53 | 34.53 | 38.48 | 46.36 | 55.77 | 34.55 | 38.53 | 46.30 | 55.95 | 34.75 | 38.72 | 46.47 |
| **Sentiment** | DIRE with Sentiment | 56.15 | 34.82 | 38.86 | 46.56 | 56.32 | 34.81 | 38.94 | 46.72 | 55.70 | 34.21 | 38.11 | 46.00 |
| | **Improvement (%) over DIRE** | 1.12% | 0.84% | 0.99% | 0.43% | 0.99% | 0.75% | 1.06% | 0.91% | -0.45% | -1.55% | -1.58% | -1.01% |

The results, presented in Table 5, reveal that the inclusion of sentiment labels led to slight performance improvements in the NAML and NRMS models. Interestingly, the FastFormer model did not benefit from the addition of sentiment labels, showing slight declines in all metrics. This outcome suggests that the FastFormer model's architecture might not leverage sentiment information as effectively as NAML and NRMS, possibly due to its different approach to handling contextual information.

Overall, the incorporation of sentiment as an extra feature demonstrates a modest enhancement in performance for the NAML and NRMS models, supporting that enriching the data can be beneficial, and it could be worth including even more features of the EB-NeRD dataset in future work. However, the varying impact across different models highlights the importance of model-specific feature integration strategies.

## 5 Conclusion, Limitations and Future Work

As discussed earlier, Liu et al. (2024) aimed to enhance content-based recommendation systems by addressing the limitations of previous methods that used CNNs and pre-trained word representations as content encoders. These early methods struggled with word-level encoding and lacked the necessary knowledge to effectively grasp content. To overcome these issues, Liu et al. (2024) adopted pretrained language models based on transformer architecture, which offer extensive information storage and can process datasets containing trillions of tokens. The authors developed the ONCE framework by integrating both open-source and closed-source LLMs. When utilizing the open-source LLMs , as stated earlier, the authors employed a discriminative recommendation. This was achieved through the replacement of the original content encoder with a LLM that ultimately enhanced user modeling and content understanding.

When replicating and conducting our experiments, we successfully managed to employ and adjust the DIRE framework to the Eb-NeRD dataset. Through several processing, tokenization, training procedures that were extracted from the original pipeline of Liu et al. (2024), table 2 and 3 were constructed. One noteworthy

difference is the resource limitation, due to the fact, that all LLaMA -based experiments of the original paper were conducted on a device with 80GB whereas our team only 40GB worth of storage allocated. This being said, we were only able to make use of LLaMA-7B open-source LLM instead of LLaMA-13B. Out of the 3 main news recommendation models which were used during our training and testing phase, the FastFormer model Wu et al. (2021) managed to perform best. This can be attributed to its general architecture which applies additive attention to handle long sequences. When leveraging closed-source LLMs, the authors employed a generative approach, named GENRE, which involves creating prompts to generate new data, updating existing data, and training models with the generated data to enhance the performance of recommendation systems. For GENRE, we successfully replicated the data augmentation, shown in 3.3 but excluded the Personalized Content Generator (CG) and the optional Chain-based Generation augmentation. This decision was due to the EB-NeRD dataset containing only 2% cold users compared to the 20% in the MIND dataset, which the original paper used. We prioritized the MIND dataset since it is also a news recommendation dataset, and the augmentation did not significantly benefit the EB-NeRD data. The repository lacked necessary files for Chain-based Generation, resulting in the omission of this component as well. On top of that, we could only replicate the data augmentation aspect of the GENRE framework, as their GitHub repository included code files solely for data generation and lacked the training and testing components for the specific models.

Incorporating sentiment as an additional feature demonstrated modest improvements in performance for the NAML and NRMS models. This finding aligns with the hypothesis that enriching the dataset with more contextual information, such as sentiment, can help the recommendation system better capture the nuances of user preferences and content characteristics. Adding further features of the EB-NeRD dataset seems like a valuable addition for future work.

Our replication of the ONCE framework, which combines open-source and closed-source LLMs, did not lead to the expected overall improvement in performance. While the DIRE framework showed positive results with the integration of open-source LLMs like LLaMA-7B, the additional tokenization and data augmentation steps in the GENRE framework did not translate into performance gains. We attribute this outcome to potential issues in the effective integration of the newly generated data into the training process, a challenge noted by other researchers in the field.

The primary limitations of our study include the computational cost of employing LLMs and the implicit biases inherent in these models. For example, ChatGPT models show biases towards data from the United States, which might have impeded performance on the Danish EB-NeRD dataset. Additionally, there are privacy concerns associated with using closed-source models, which may restrict their application in certain contexts. Future research could explore a hybrid approach that integrates LLMs with traditional recommendation algorithms to improve computational efficiency and balance model biases. Moving beyond the scope of the current implementation, future work may also explore adding a real-time user feedback mechanisms could enhance the personalization of recommendations, making for a more agile system that adapts dynamically to user preferences over time.

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# A Appendix 1

## A.1 Ablation Study on Open-source LLMs

On top of the main experiments we conducted when trying to replicate the original paper, we also investigated the impact of several parameters on the overall performance of the different news recommendation models when using an Open-source LLMs (LlaMA-7B). More specifically, we tested various batch sizes alongside its accumulation batch size. The accumulation batch size was responsible for controlling the gradient accumulation which directly impacts the training stability, performance and total number of samples for each back propagation. Moreover, we also investigated the impact of various learning rates, number of layers, and used item attributes. When trying to utilize and inspect layers of Llama-7b (31,30,29,27), we unfortunately were able to only use layer 31 since the other ones caused out-of-memory errors which prevented us from further inspecting the cause or solution. When inspecting the impact of used item attributes, the original pipeline initially had title and category as attributes. We therefore then decided to also include subtitle and body as additional attributes.

Table 5: Performance comparison of various batch sizes on the EB-NeRD dataset.

| | **Model** | Fastformer (2021b) | | | |
|---|---|---|---|---|---|
| | **Metric** | AUC | MRR | N@5 | N@10 |
| **DIRE-LLaMA7B (Ours)** | Batch size(16) & acc batch size (4) | 55.94 | 34.65 | 38.69 | 46.44 |
| | Batch size(64) & acc batch size (1) | 56.40 | **35.03** | **39.02** | **46.77** |
| | Batch size(32) & acc batch size (2) | **55.95** | 34.75 | 38.72 | 46.47 |

Table 6: Performance comparison of various learning rates on the EB-NeRD dataset.

| | **Model** | NRMS (2019c) | | | | Fastformer (2021b) | | | |
|---|---|---|---|---|---|---|---|---|---|
| | **Metric** | AUC | MRR | N@5 | N@10 | AUC | MRR | N@5 | N@10 |
| **DIRE-LLaMA7B (Ours)** | lr = 0.001 | 56.26 | 34.82 | 38.79 | 46.56 | 55.74 | 34.58 | 38.55 | 46.29 |
| | lr = 0.0001 | 55.77 | 34.55 | 38.53 | 46.30 | 55.74* | 34.58* | 38.55* | 46.29* |
| | lr = 0.0005 | 56.26* | 34.82* | 38.79* | 46.56* | 55.74* | 34.58* | 38.55* | 46.29* |

* The values for Fastformer at learning rates 0.0001 and 0.0005 are the same as those at 0.001.
* The values for NRMS at learning rate 0.0005 are the as those at 0.001.

When referring to table 5 , using a higher batch size proved to make a positive impact since 3/4 metrics of the second row were dominating the other metric values from smaller batch sizes. While smaller batch sizes provide more frequent updates, larger ones exchange frequency with stability which evidently is an advantage in news recommendation models. Other larger values of batch sizes could be utilized if we would have more storage and time allocated to our team. Therefore, a more diverse and larger set of batch sizes is advised for future work and inspection. To observe the impact of adding more user item attributes, we can still refer to table 5. More specifically, the last row consisting of a batch size (32) with accumulated batch size (2) produces the identical metric values found when using the following item attributes (title, category, subtitle, body). This implies that adding more item attributes (on top of already existing ones : title, category) does not necessarily prove to be an advantage or impactful factor when wanting to enhance the performance of the recommendation models.

When inspecting table 6, the impact of various learning rates has been documented. When taking a closer look at the Fastformer metric values, the learning rate hyper-parameter does not have that much of an impact. This is due to the fact that identical values have been found throughout three different learning rate values which imply that using the default learning rate value (0.001) is sufficient enough. When inspecting the NRMS model, on the other hand, there is a slight difference in performance between learning rate (0.001) and (0.0001), however, the higher metric values of learning rate (0.001) dominate over the other learning rate values which also show that using the default learning rate value of the original paper framework is sufficient.

## B Appendix 2